

Sergei Kliuikov

DEVOPS ENGINEER

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DevOps Engineer and Infrastructure Lead with 11 years of experience in **AWS**, **Kubernetes**, enterprise-scale cloud architectures and automation. **Core architect of high-load infrastructures** (1TB+/hour, 2B+ transactions) and **cloud-native solutions** (1,000+ production servers). Created infrastructure automation tools with **600+ GitHub stars**. Expert in achieving **five-nines reliability** (99.999%) for mission-critical systems serving **77M+** users.

- **VST Consulting:** IT consulting company with **100+ projects** and **2 patented innovations** that helped elevate a client's valuation from **\$150M** to **\$1.4B**.
- **PAO Megafon:** A major telecom operator serving **77.26 million subscribers** nationwide, managing over **1,000 HLS servers** and **50+ high-load billing servers**
- **Innovative Utilities Systems:** An IT solutions provider managing integrated database systems for residential complexes, handling **500+ concurrent users** and processing **10TB+ of operational data** across distributed infrastructure.

KEY ACHIEVEMENTS

- **Led enterprise cloud transformation** for Fortune 500 client (\$1.4B acquisition), architecting infrastructure for **200+ microservices** with automated deployment pipelines. Implemented **zero-downtime deployment strategy** and achieved **99.99% system reliability** across multiple regions.
- **Designed distributed logging infrastructure** scaling to **1TB/hour** across 3 data centers. Implemented custom retention policies and compression algorithms, reducing storage costs by **\$500K annually** while enabling real-time incident detection and cutting MTTR from 4 hours to 30 minutes.
- **Created Polemarch** - infrastructure-as-code orchestration platform (**600+ GitHub stars**) managing **1,000+ servers**. Platform enabled **40% faster deployments** and standardized automation practices across enterprise clients. System became standard solution for **20+ Fortune 500 companies**.
- **Built high-availability infrastructure** with geo-distributed failover handling **50K+ requests/second**. Implemented automated scaling and self-healing capabilities achieving **99.999% uptime** for mission-critical services across multiple regions.
- **Established DevOps practice** growing from 3 to 10 engineers. Implemented **GitOps workflows**, **CI/CD pipelines**, and **Infrastructure as Code** practices, resulting in **5x faster deployments** and **80% reduction** in production incidents.

Expertise in **system programming** (Cython, Python C-API, Rust PyO3, POSIX), **distributed systems** (processing 1TB+/hour, 10000+msg/sec), **low-level optimization** (2x performance improvements, PostgreSQL query plans enchantments), and **open-source development** (core contributor to Django REST Framework, drf-yasg and Django-Elasticsearch, contributor to ormsgpack, uWSGI, flower, daphne, pycard). Advanced knowledge of **cloud-native architectures** (Kubernetes operators, custom Helm controllers) and **high-load infrastructure** (1,000+ server clusters).

Authorized to work for any U.S. employer without visa sponsorship.

PROFESSIONAL EXPERIENCE

VST CONSULTING

May 2015 – January 2025

*Technology consulting company delivering cloud infrastructure solutions for Fortune 500 clients, managing **1,000+ production servers** and achieving **99.999% uptime** for mission-critical systems.*

Senior DevOps

- **Architected distributed log management system** processing **1TB/hour across 3 data centers**. Implemented custom retention policies and compression algorithms, reducing storage costs by 75% while maintaining sub-second query performance. **Led team of 10 engineers**.
- **Developed high-availability DNS infrastructure** with GEO-based synchronization serving **50K+ requests/second**. Custom solution enabled automatic failover across regions with **99.999% uptime** for mission-critical services.
- **Re-engineered cloud infrastructure** by optimizing load distribution between hypervisors - **cutting costs by 50%**.
- **Created Polemarch** - low-level infrastructure orchestration platform (**600+ GitHub stars**) integrating with **kernel-level operations**. System manages **1,000+ servers** across multiple data centers, handling complex deployment scenarios.
- **Built configparserC** - high-performance configuration parser in C/Cython achieving **3x speed improvement** over Python implementations. Parser handles **100K+ configurations/second** in production environments.
- **Developed network intrusion detection system** using Snort and custom kernel modules, processing **10Gb/s** of network traffic with real-time analysis and threat detection.
- **Automated new client deployments** to reduce provisioning times 10-fold, thereby **enabling to serve two orders** of magnitude more clients.
- **Engineered bare-metal clusters** and network infrastructure for development and production environments to **reduce infrastructure costs by 10x over two years**.
- **Built technical department from ground up**, growing team from 3 to 10 engineers across 2 development offices. **Established modern engineering practices** (DevOps, TDD, Gitflow) and mentorship program, resulting in **15x faster release cycles**, **80% reduction in production incidents**, and successful delivery of 15+ enterprise-scale projects. Team became company's primary innovation hub, contributing to both patented solutions.

PAO MEGAFON

May 2014 – May 2015

Major telecom operator with infrastructure of **1,000+ HLS servers** and **50+ high-load billing servers**, processing **2B+ monthly transactions** with **99.95% SLA**

DevOps

- **Led 5-person critical systems team** managing infrastructure for **77.26M subscribers**. Implemented **real-time monitoring system with pattern-based anomaly detection**, reducing critical incidents by 70% across **1,000+ streaming servers**.
- **Automated operational tasks** for large-scale cluster of **1,000+ HLS servers** and **50+ billing servers**, reducing maintenance effort by **120+ engineer hours monthly** and cutting operational costs by **\$25K/month**. System handled **automated failover**, **capacity planning**, and **resource optimization** across regional data centers.
- **Engineered low-level Linux debugging toolset** for critical system analysis, enabling deep kernel-level diagnostics across billing infrastructure. Tool reduced critical incident resolution time from **6 hours to 30 minutes** and became standard solution for **emergency response team across 5 regions**, saving **\$100K+ in potential revenue loss** per incident.
- **Improved billing platform reliability from 99.8% to 99.95% SLA** by implementing custom load balancing solution and automated failover mechanisms. System handles **2B+ transactions monthly** with zero data loss, supporting **critical financial operations across 77M+ subscribers**.

INNOVATIVE UTILITIES SYSTEMS

November 2013 – April 2014

Built and managed distributed infrastructure processing **10TB+ of operational data** across multi-datacenter environment with **real-time processing capabilities**.

Software Engineer

- **Built high-performance database system** serving **1,000+ residential units** and **200+ internal developers**. Implemented custom caching layer and query optimization, reducing average response time from **2s to 200ms** under peak loads.
- **Developed secure authentication system** with role-based access control and encryption for sensitive billing data. Solution detected and prevented **100+ unauthorized access attempts** monthly while maintaining sub-second response times.
- **Created real-time integration pipeline** between billing and monitoring systems processing **50K+ daily transactions**. Automated data synchronization reduced manual reconciliation effort from 4 hours to 15 minutes daily.

EDUCATION

Bachelor's Degree in Applied Computer Science, Moscow Technological Institute

PROFESSIONAL DEVELOPMENT

Certified Kubernetes Administrator, CNCF

May 2024

COMMUNITY LEADERSHIP

- **Core contributor** to Django REST Framework, drf-yasg and Django-Elasticsearch
- Created **ConfigparserC** - high-performance Cython parser with **3x speed improvement**
- Built **Kubernetes operator** for automated deployments used by **20+ teams**
- **Co-organized DevOps Days** conference, growing attendance from 30 to 80 participants
- Developed universal Helm chart adopted by **15+ enterprise projects**
- Developed ansible role for package installation with 3000+ downloads

EXECUTIVE STRENGTHS

Team Leadership, Agile Management, Software Architecture, System Integration, DevOps, Continuous Delivery, Deployment Automation, Software Configuration Management, Test-Driven Development (TDD), Strategic Innovation, Process Optimization, Problem Solving, Cross-Functional Collaboration, Open-Source Community Leadership, Mentorship, Coaching, Stakeholder Communication

TECHNICAL SKILLS

Python, Cython, Rust (PyO3), Go, php, C, C#, SQL, XML, HTML, CSS, bash, Django, Django Rest Framework, FastAPI, Flask, Celery, Typescript, Javascript, Vue, Vuetify, Nuxt, Bootstrap, Capacitor, Node.js, OpenAI, Kubernetes, Kubernetes Operator, kubescape, k3s, talos, metallb, werf, Docker, dive, Ansible, ArgoCD, Helm, Rancher, GitLab CI/CD, GitHub Actions, Linux, Bash, Ansible, Chef, CloudFormation, CloudFront, AWS, OpenStack, Proxmox, TrueNAS, Postgres, MySQL, SQLite, Redis, tarantool, Centrifugo, ELK Stack, RabbitMQ, Nginx, Traefik, HAProxy, Grafana, VictoriaMetrics, Minio, TCP/IP, REST API, OpenAPI, OAuth2, JWT, ORM, Git, Gitflow, GitOps, server management, virtualization, containerization